



Laxmi Charitable Trust's

Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce

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Department of Information Technology (Bsc.IT Semester III)

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Python for DS Project Report

1. Title of Project:-MOVIES DATA ANALYSIS AND DASHBOARD

Tools Used: Python, NumPy, Pandas, Matplotlib, Seaborn, Beautiful Soup

2. Introduction

Movies data analysis helps us understand movie titles, release years, genres, actors, directors, sequel information, and hit or flop status. In this project, a movie dataset containing movie records is analyzed using Python libraries. Different statistical methods and visualizations are used to identify patterns and present important results through a dashboard.

3. Objective

- To analyze movies data using Python.
- To clean and organize the dataset using Pandas.
- To perform numerical and statistical analysis using NumPy and Pandas.
- To create meaningful visualizations using Matplotlib and Seaborn.
- To understand movie release trends, genres, and hit or flop status.
- To present important results using a **Movies Data Analysis Dashboard**.
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4. Dataset Description

The Movies dataset contains information about movies, including their identification, title, release information, genre, writers, actors, directors, sequel information, and hit or flop status .

Column	Description
imdbId	The unique IMDb identification number of the movie.
title	The name or title of the movie.
releaseYear	The year in which the movie was released.
releaseDate	The date on which the movie was released.
genre	The genre or category of the movie.
writers	The writers associated with the movie.
actors	The actors who participated in the movie.
directors	The directors of the movie.
sequel	Indicates whether the movie is a sequel.
hitFlop	Represents the hit or flop status of the movie.

5. Tools and Libraries Used

Library	Purpose
NumPy	Numerical calculations and array operations
Pandas	Dataset loading, cleaning, and analysis
Matplotlib	Data visualization
Seaborn	Statistical visualization
Beautiful Soup	Extracting data from HTML/web pages

6. Data Cleaning

Cleaning Operations Performed

1. Removed duplicate movie records using `drop_duplicates()`.
2. Converted the `releaseYear` column into numeric format.
3. Converted the `releaseDate` column into proper date format.
4. Handled missing text values using "Unknown".
5. Handled missing numerical values using the median where required.
6. Removed extra spaces from column names using `str.strip()`.
7. Removed unnecessary spaces from text values.

Data Cleaning

```
df.columns = df.columns.str.strip()
```

```
df = df.drop_duplicates()
```

```
df["releaseYear"] = pd.to_numeric(  
    df["releaseYear"],  
    errors="coerce"  
)
```

```
df["releaseDate"] = pd.to_datetime(  
    df["releaseDate"],  
    errors="coerce"  
)
```

```
text_columns = [  
    "title",  
    "genre",  
    "writers",  
    "actors",  
    "directors",  
    "hitFlop"  
]
```

```
for col in text_columns:  
    df[col] = df[col].fillna("Unknown")
```

```

df[col] = df[col].astype(str).str.strip()

df["releaseYear"] = df["releaseYear"].fillna(
    df["releaseYear"].median()
)

print("Data Cleaning Completed!")

print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

print("\nDuplicate Records After Cleaning:")
print(df.duplicated().sum())

```

Output:-

```

imdbId      0
title       0
releaseYear  0
releaseDate 53
genre       2
writers     119
actors      3
directors   4
sequel      3
hitFlop     0
dtype: int64

```



7. Data Analysis

Code:-

```

# Data Analysis

total_movies = df["title"].count()

average_release_year = np.mean(
    df["releaseYear"]
)

```

)

```
maximum_release_year = np.max(  
    df["releaseYear"]  
)
```

```
minimum_release_year = np.min(  
    df["releaseYear"]  
)
```

```
genre_count = df["genre"].value_counts()
```

```
hit_flop_count = df["hitFlop"].value_counts()
```

```
year_count = df["releaseYear"].value_counts()
```

```
director_count = df["directors"].value_counts()
```

```
print("MOVIES DATA ANALYSIS RESULTS")  
print("-----")
```

```
print(  
    "Total Movies:",  
    total_movies  
)
```

```
print(  
    "Average Release Year:",  
    round(average_release_year, 2)  
)
```

```
print(  
    "Latest Release Year:",  
    maximum_release_year  
)
```

```
print(  
    "Earliest Release Year:",  
    minimum_release_year
```

)

```
print("\nMovie Count by Genre:")  
print(genre_count)
```

```
print("\nHit/Flop Count:")  
print(hit_flop_count)
```

```
print("\nMovies by Release Year:")  
print(year_count)
```

```
print("\nMost Frequent Directors:")  
print(director_count.head(10))
```

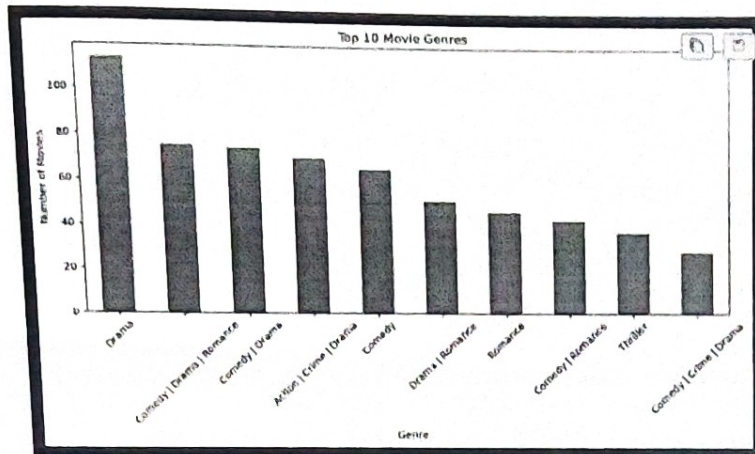
8. Data Visualizations

A. Bar chart

Code:-

```
plt.figure(figsize=(10, 6))  
  
df["genre"].value_counts().head(10).plot(kind="bar")  
  
plt.title("Top 10 Movie Genres")  
plt.xlabel("Genre")  
plt.ylabel("Number of Movies")  
plt.xticks(rotation=45)  
plt.tight_layout()  
plt.show()
```

Output:-



B. Line chart

Code:-

```
plt.figure(figsize=(10, 6))
```

```
df["releaseYear"].value_counts().sort_index().plot(kind="line")
```

```
plt.title("Movies by Release Year")
```

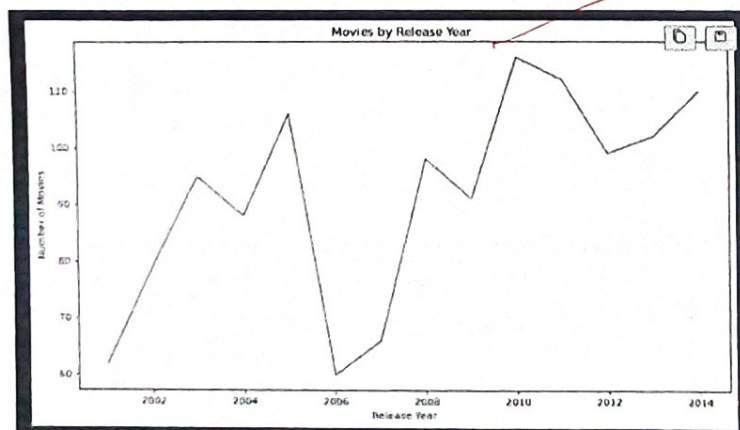
```
plt.xlabel("Release Year")
```

```
plt.ylabel("Number of Movies")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:-



C. Histogram

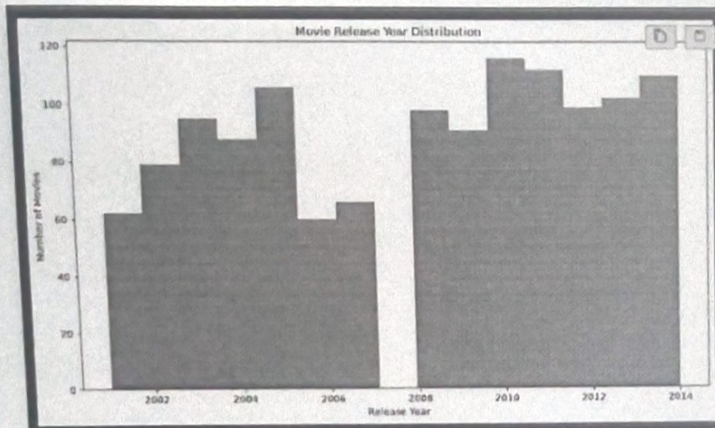
Code:-

```
plt.figure(figsize=(10, 6))
```

```
plt.hist(df["releaseYear"], bins=15)
```

```
plt.title("Movie Release Year Distribution")
plt.xlabel("Release Year")
plt.ylabel("Number of Movies")
plt.tight_layout()
plt.show()
```

Output:-



D. Box Plot –Movie Release Year box Plot

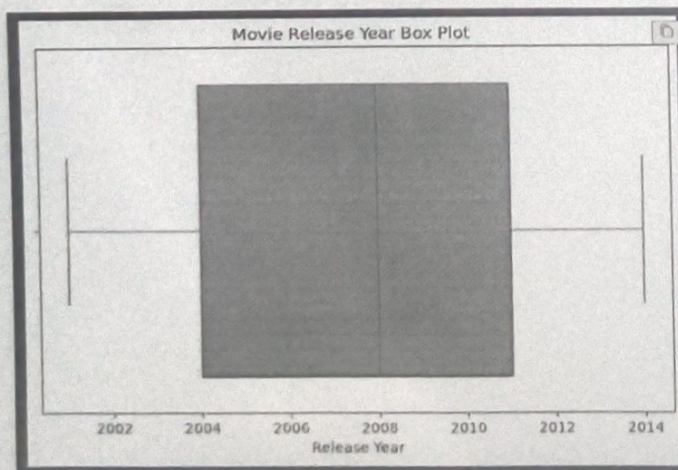
Code:-

```
plt.figure(figsize=(8, 5))

sns.boxplot(x=df["releaseYear"])

plt.title("Movie Release Year Box Plot")
plt.xlabel("Release Year")
plt.show()
```

Output:-



E. Pie Chart– Sequel Distribution

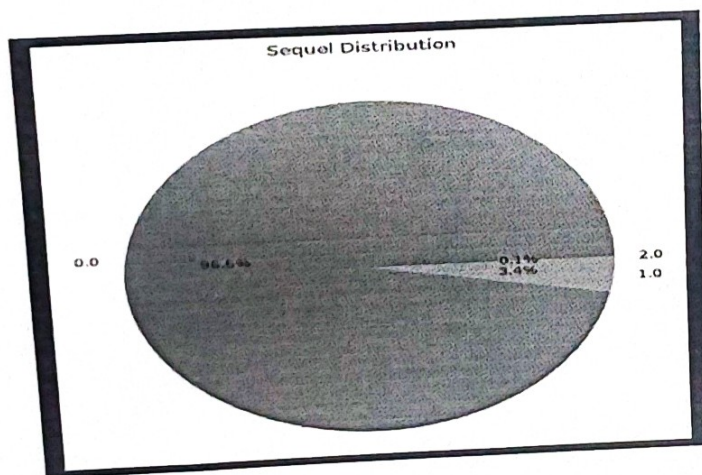
Code:-

```
plt.figure(figsize=(7, 7))

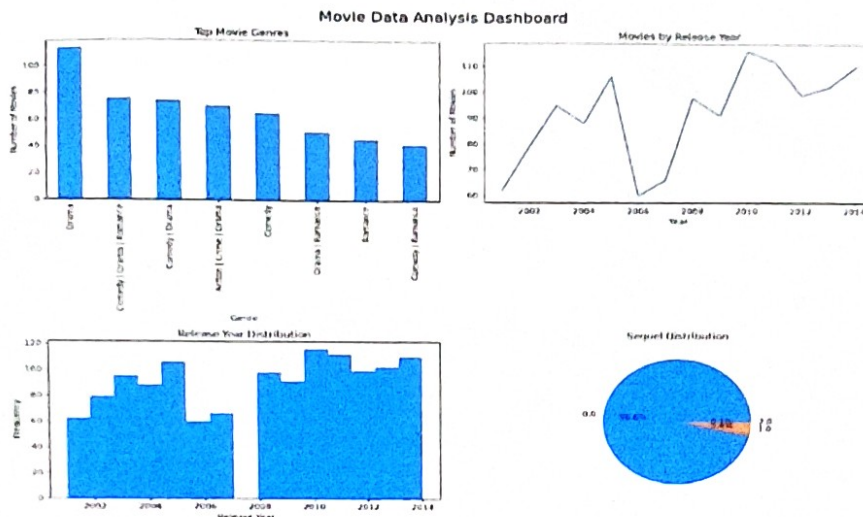
df["sequel"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%"
)

plt.title("Sequel Distribution")
plt.ylabel("")
plt.show()
```

Output:-



9. Dashboard



10. Key Findings/Observations

1. The dataset contains movie records with information about movie titles, release years, genres, actors, directors, and hit or flop status.
2. The analysis shows the distribution of movies across different genres.
3. The release-year analysis helps identify the years in which more movies were released.
4. The genre-wise analysis shows which genres are most frequently represented in the dataset.
5. The hit and flop analysis shows the distribution of movies according to their performance status.

11. Conclusion

The Movies Data Analysis and Dashboard project successfully analyzed movie data using Python. Pandas and NumPy were used for data loading, cleaning, and analysis, while Matplotlib and Seaborn were used to create meaningful visualizations. The analysis helped identify movie release patterns, genre distribution, and hit or flop status. BeautifulSoup can be used for basic web data collection from publicly available webpage content. The final dashboard presents important statistics, visualizations, and findings in an easy-to-understand format.

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